# Departmental Expense Management System (DEMS)

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# 1. Project Kickoff

**Project Goals:**
The goal of this project is to design and implement a Departmental Expense Management System (DEMS) that allows centralized tracking and analysis of departmental spending. The system will feature a structured relational database, an ETL pipeline to import real-world-style expense data, and interactive dashboards for insights.

**Project Scope:**
The scope includes:

- Designing a normalized schema using PostgreSQL.
- Implementing a Python-based ETL pipeline.
- Developing dashboards for budget vs actual tracking.
- Applying basic data governance concepts.

It excludes predictive analytics, machine learning, or real-time feeds to prevent scope creep.

**Deliverables & Milestones:**

1. Weeks 2–3: Schema design and ER diagram.
2. Weeks 4–5: SQL database setup and data import.
3. Weeks 6–7: ETL pipeline and data integration.
4. Week 8: Dashboard development.
5. Week 9: Testing and documentation.
6. Week 10: Final submission.

**Timeline and Dataset:**
The dataset is simulated and includes fields like department, vendor, date, category, and amount. It is stored in CSV and uploaded to GitHub under `/data/sample_expense_data.csv`.

**Readiness and Gaps:**
As a solo contributor with a finance background and intermediate SQL/Python skills, I am well-prepared for this project. Minor gaps in Tableau will be addressed by optionally using Dash for dashboard creation.

# 2. Team Discussions

**Team Composition:**
This is an individual project.

**Core Skills:**

- SQL and ER modeling

- Financial logic and domain knowledge
- Python scripting (ETL with Pandas)
- Basic dashboard creation (Dash/Tableau)

**Programming Tools and Platforms:**

- PostgreSQL
- Python (Pandas, SQLAlchemy)
- Dash (or Tableau)
- GitHub and Overleaf

# 3. Skills and Tools Assessment

**Tools and Frameworks:**

- **Database:** PostgreSQL
- **ETL:** Python with Pandas
- **Visualization:** Dash, optionally Tableau
- **Documentation:** Overleaf (LaTeX)
- **Version Control:** GitHub

**External Resources:**

- Tableau documentation
- Dash tutorials
- PostgreSQL schema examples

**Task Assignment:**
As a solo developer, I am handling all components. Milestones and progress are tracked in the Excel tracker file.

# 4. Submission for This Iteration

**Tasks Completed:**

- Project idea selection and scoping
- Drafting of ER diagram
- Sample dataset generation
- GitHub repository setup
- Excel progress tracker completed

**Challenges:**

- Tableau learning curve (mitigated with Dash)
- GitHub versioning setup (resolved via guides)

**Data Link:**
CSV file hosted at: `/data/sample_expense_data.csv` in the GitHub repo.

**Excel Tracker:**
Tracker file is named `project_tracker.xlsx` and included in the repository.

**ER Diagram:**

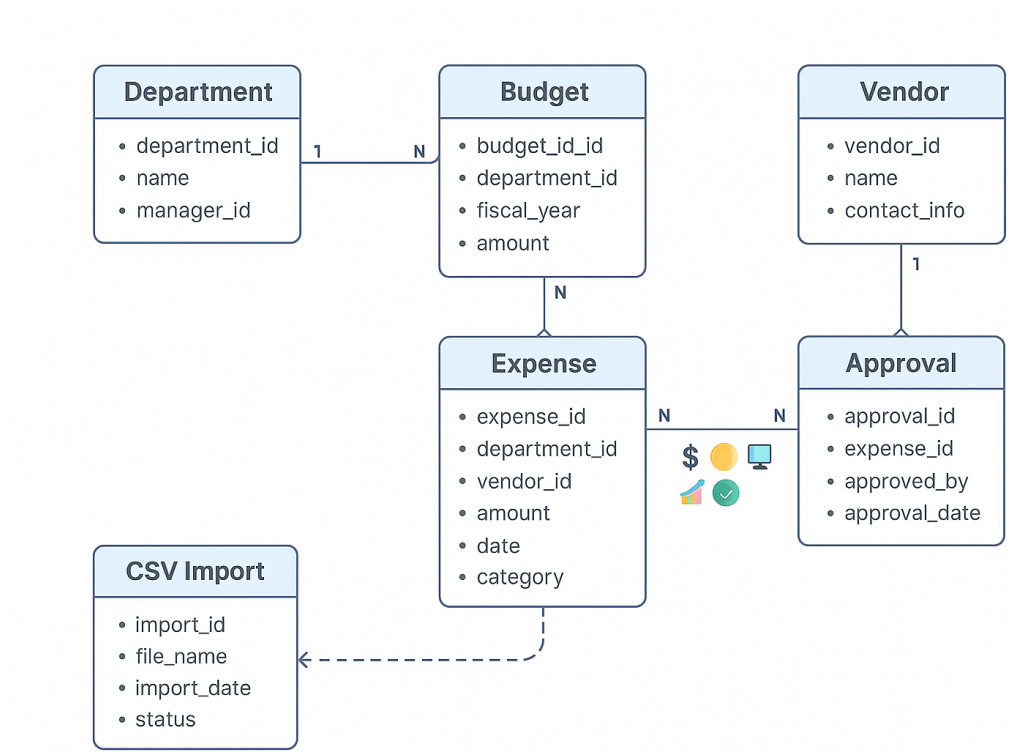


Figure 1: Entity Relationship Diagram for DEMS

**GitHub Repository:**
`https://github.com/shivxmpxtel21/Department-Expense-Management-System`